

Ideas



World map
On obesity
(choropleth map)



Bar chart on
BMI
(countries with
highest BMI)



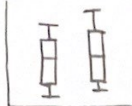
Line chart on
obesity rate
in Australia



Mean BMI
Bubble plot for
different genders



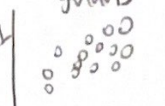
Mean BMI
Scatter plot for
different genders



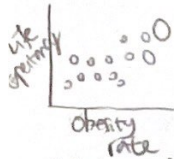
Box and whisker
plot for
worldwide obesity
rates



Daily calorie
bubble plot for
calorie intake vs
obesity rate



Daily calorie
bubble plot for
calorie intake vs
BMI



Bubble plot for
obesity rate
vs. life expectancy



Country
Lollipop chart on
obesity rate



Line chart on
BMI in
Australia



side by side
bar chart for
obesity rate and
BMI (for continents)



stacked bar chart on weight status in Australia

Combine and refine



Bar chart on
BMI



Tooltip

The BMI information
can be included on the
tooltip of the world map.

Filter



Mean BMI
Scatter plot for
different genders



Bar chart on
BMI

Similar to bubble plot
Bubble plot can encode
population using
size of circles.

Lollipop chart is
a better alternative,
especially when
many countries have
similar average BMI.

Categorize

Overview of obesity rate
worldwide



obesity rate in
Australia



BMI and obesity rate



Daily calorie and obesity
rate

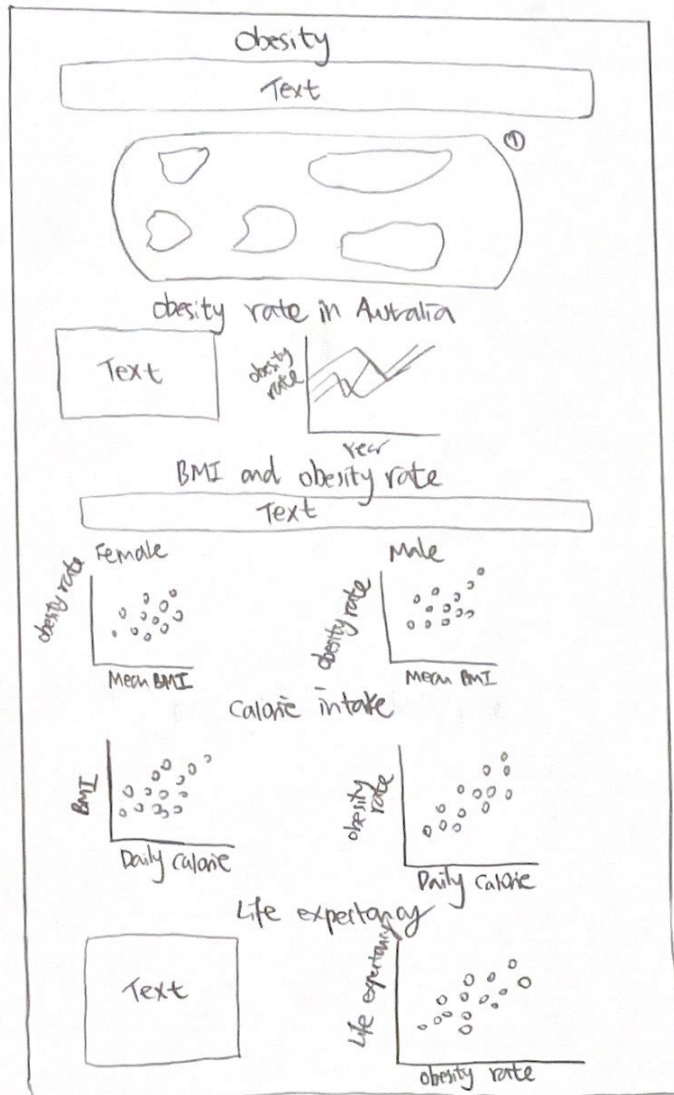


Ramifications of
high obesity rates

Question

- Are there better alternatives to bubble plot?
- What would be a good layout?
- What conclusions can be drawn from these plots?

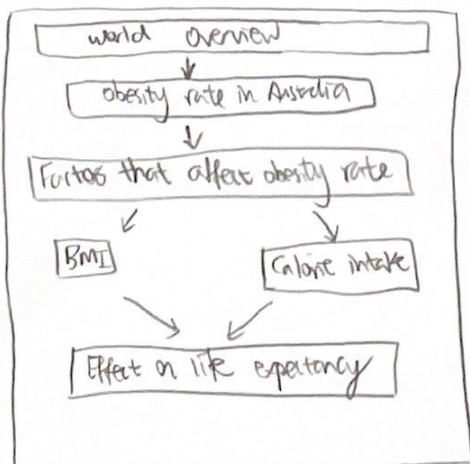
Layout



①: world map (choropleth map or obesity rate)

Focus

no key focus. Every component is important.
General flow:



Title: Dashboard on obesity Design #1

Author: Yitang zheng

Date: 4/10/2025

Sheet: 2

Task: Design an informative dashboard

operation

For the bubble charts there could be a drop-down menu for continent selection.

There can also be sliders to show countries with BMI / obesity rate / life expectancy above a certain threshold.

Discussion

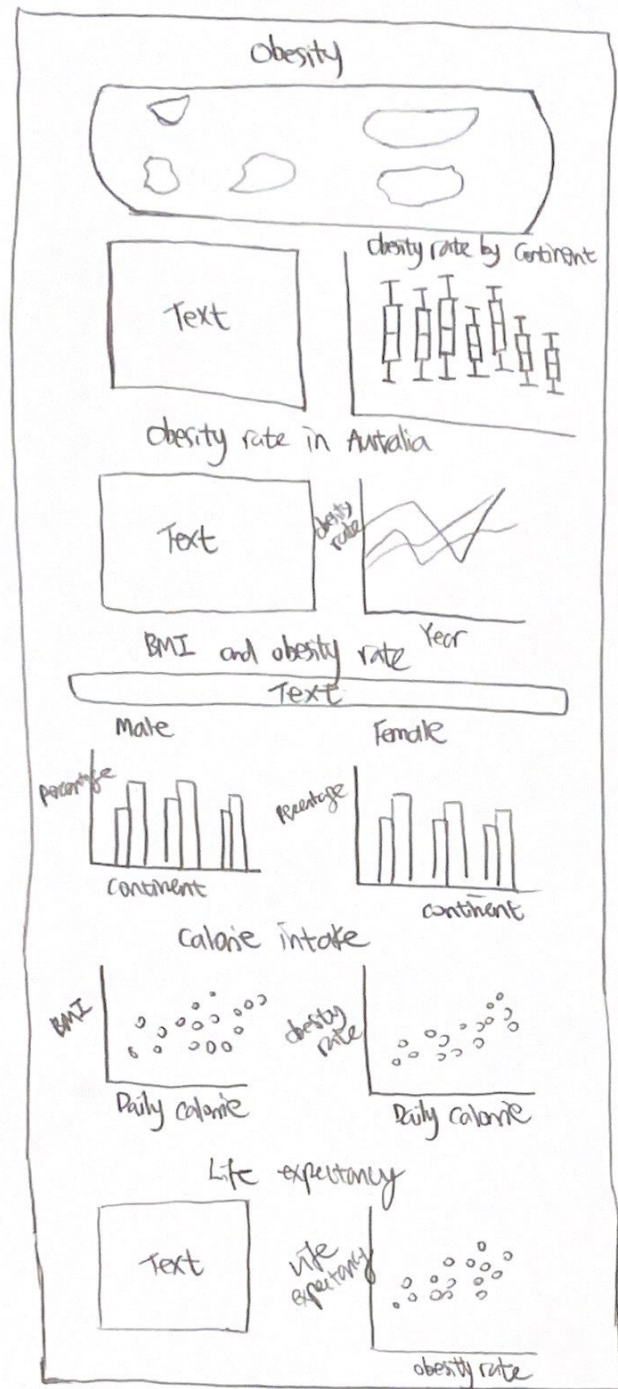
Pros

- good layout - mostly two columns approach
- good degree of interactivity

Cons

- limited types of idioms, could consider changing the idioms for some of the bubble charts.

Layout



Focus

- no key focus.
- same general flow as design #1.

Title: Dashboard on obesity Design #2

Author: Xinyi Zhang

Date: 5/10/2025

Sheet: 3

Task: Design an informative dashboard

operation

For the line chart on obesity rate in Australia, there can be selection interaction, where you can highlight a line and the other lines have lower opacity.

The bubble charts will have the same interactions (the drop-down menu and slider) as Design #1.

Discussion

Pros

- A greater variety of idioms compared to design #1
- Integrates a by continent approach. This may highlight interesting findings that will not be covered in the previous design.

Cons

- There may not be enough interactions

Layout



Focus

no key focus.

similar general flow as design #2, except how the calorie intake section was removed.

Title: Dashboard on obesity Design #3

Author: Yitong zheng

Date: 5/10/2025

Sheet: 4

Task: Design an informative dashboard

operation

①: For the line chart, the users can select which line they want to see using the drop-down menu.

②: There can be two interactions for the BMI and obesity bubble charts. The users can select if they want to see the bubble chart for female or male. there will also be a slider that filters by population.

Discussion

Pros

- Has more interactions than all previous designs
- more concise
- consistent two-column layout

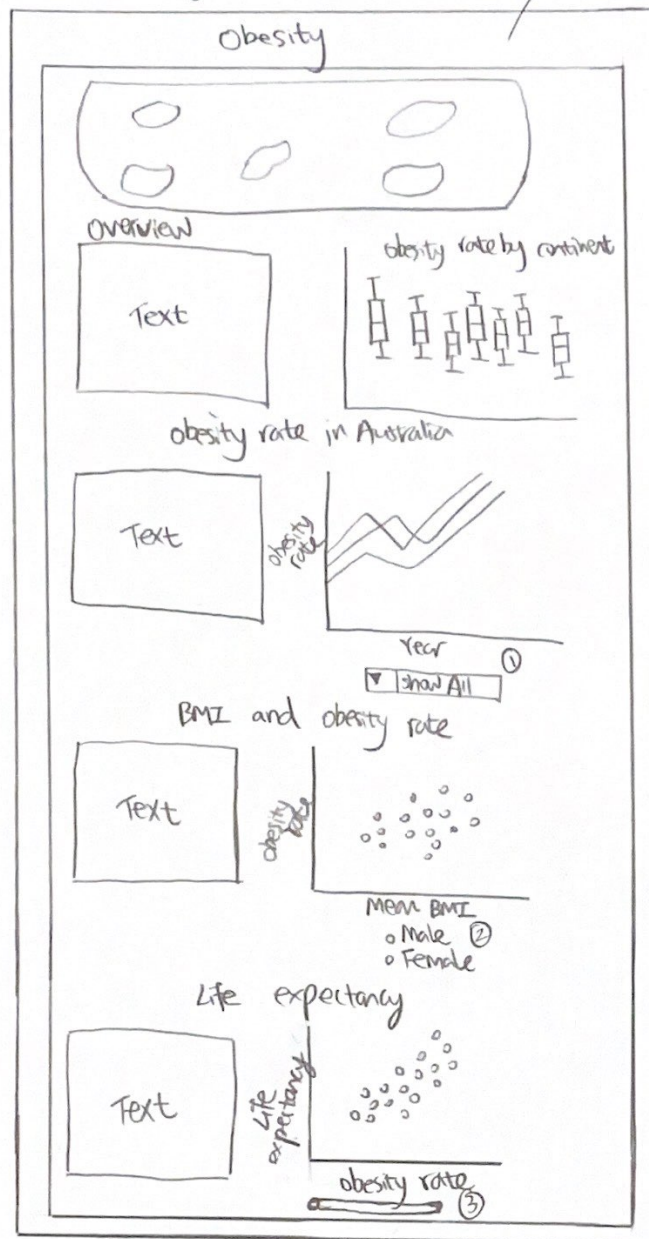
cons

- Less number of idioms.

Layout

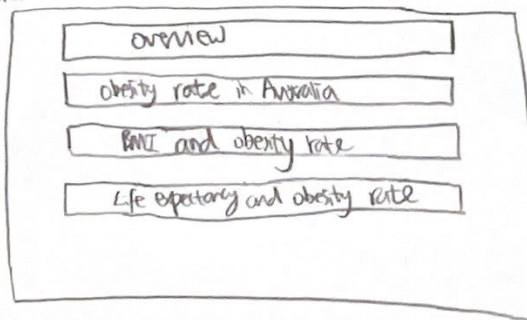
Based on design #3.

Background



Focus

All sections are equally dominant.
General Flow:



Title: The final design

Author: Xitang zhong

Date: 6/10/2025

Sheet: 5

Task: Finalise the design of the dashboard

operation

- ①: There is a drop-down menu for the line chart. The default would be 'show All' where all lines are shown, but the users can also choose to see each line separately.
- ②: For the bubble charts on BMI and obesity rate, there would be buttons to allow the users to switch between the bubble charts on male and female.
- ③: For the bubble chart on obesity rate and life expectancy, the users can filter by minimum population.

Detail

- Python will be used for all data wrangling.
- webpage will be built using html, css and Vega-Lite.
- Font from Google Font:
 - Use a serif font for heading and sub-headings.
 - Use a sans-serif font for the main text.